

Packages in Java

A package in Java is a namespace that groups related classes and interfaces together. Packages are used to organize code, avoid naming conflicts, and control access.

Why Packages are Needed

- Organize large applications into logical groups
- Avoid class name conflicts
- Support access protection
- Improve code reusability and maintenance

Types of Packages

1. **Built-in Packages:** Provided by Java API (java.lang, java.util, java.io, java.sql).
2. **User-defined Packages:** Created by programmers to organize their own classes.

Creating a Package

Use the package keyword as the first line of the source file:
`package com.company.project;`

Importing Packages

- `import java.util.Scanner;`
- `import java.util.*;`
- `java.util.Scanner sc = new java.util.Scanner(System.in);`

Packages and Access Modifiers

Modifier	Same Package	Different Package
public	Yes	Yes
protected	Yes	Yes (subclass only)
default	Yes	No
private	No	No

Interview Definition

A package in Java is a namespace that organizes related classes and interfaces into a hierarchical structure, helping in code reuse, access control, and maintainability.